

Volume 18 Swarm Intelligence - Swarm Intelligence Part 1

Version v91 draft

README.md

Swarm Intelligence Part 1

Version: v91 draft

Status: Draft

Document ID: MAL-V18-V91-README

Purpose

This release folder contains the official v91 documentation set for Swarm Intelligence Part 1.

Scope

- Covers Swarm Formation.
- Covers Review Loops.

Acceptance

- All documents contain implementation-level guidance.
- The release PDF and ZIP are generated and verified.
- MANIFEST and RELEASE_NOTES are updated for v91.

Swarm Intelligence Part 1 Index

- [guide-18-91-01-swarm-formation.md](#) - Swarm Formation
- [guide-18-91-02-review-loops.md](#) - Review Loops

Swarm Formation

Title: Swarm Formation

Version: v91 draft

Status: Draft

Document ID: MAL-V18-SWARM-FORMATION-01

Purpose

Provide official implementation guidance for Swarm Formation inside Mariam AI Enterprise OS.

Scope

- Defines the recommended workflow, required evidence, roles, inputs, outputs, interfaces, and acceptance checks.
- Applies to architecture, engineering, operations, testing, governance, and release work connected to this topic.
- Excludes temporary experiments unless they are promoted into official documentation.

Responsibilities

- Owners maintain the document, examples, and acceptance evidence.
- Implementers follow the workflow and record deviations.
- Reviewers verify security, quality, traceability, and operational readiness.
- Operators confirm deployment, monitoring, recovery, and maintenance requirements where relevant.

Operating Model

- Plan the work from documented requirements and trace it to Blueprint or Specifications.
- Implement in small increments with tests, observability, and rollback behavior.
- Review changes against standards, threat model, failure modes, and acceptance evidence.
- Release only when documentation, tests, and operational notes are current.

Inputs

- Architecture Library documents, implementation tickets, configuration, schemas, policies, runtime context, and stakeholder constraints.
- Prior release notes, known risks, incident learnings, test results, and acceptance criteria.

Outputs

- Updated documentation, implementation plans, review evidence, test reports, operational runbooks, and release artifacts.
- Traceable decisions that explain why the selected approach is correct for Mariam.

Interfaces

- Git repository, CI/CD pipeline, runtime services, event bus, storage, dashboards, issue tracker, and release packaging.

- Human interfaces: review boards, chief office, department leads, operators, testers, and governance approvers.

Security

- Apply least privilege, secret isolation, signed releases where applicable, and audit-ready evidence.
- Do not expose tenant data, credentials, customer materials, private model artifacts, or privileged runtime outputs.
- Escalate security-sensitive deviations before implementation or release.

Metrics

- Lead time, review time, test pass rate, defect leakage, incident count, recovery time, documentation freshness, and acceptance completion.
- Topic-specific service metrics must be linked to dashboards and release notes.

Testing

- Verify expected behavior, failure behavior, recovery behavior, and evidence generation.
- Include automated tests where implementation exists and manual acceptance checks where documentation is the deliverable.
- Record test gaps explicitly in release notes.

Acceptance Criteria

- The workflow is clear enough for a new team member to execute without oral context.
- Required inputs, outputs, owners, interfaces, tests, and evidence are documented.
- Risks and security considerations are visible before release.
- The document is linked from INDEX, TRACEABILITY, ACCEPTANCE, MANIFEST, and release notes.

Traceability

Source	Relationship
MAL-V18-SWARM-FORMATION-01	Governs Swarm Formation.
Volume 05	Engineering standards reference.
RELEASE_NOTES_v91_draft.md	Release evidence.

Version History

Version	Date	Status	Notes
v91 draft	2026-06-29	Draft	Added official documentation for Swarm Formation.

Review Loops

Title: Review Loops

Version: v91 draft

Status: Draft

Document ID: MAL-V18-REVIEW-LOOPS-02

Purpose

Provide official implementation guidance for Review Loops inside Mariam AI Enterprise OS.

Scope

- Defines the recommended workflow, required evidence, roles, inputs, outputs, interfaces, and acceptance checks.
- Applies to architecture, engineering, operations, testing, governance, and release work connected to this topic.
- Excludes temporary experiments unless they are promoted into official documentation.

Responsibilities

- Owners maintain the document, examples, and acceptance evidence.
- Implementers follow the workflow and record deviations.
- Reviewers verify security, quality, traceability, and operational readiness.
- Operators confirm deployment, monitoring, recovery, and maintenance requirements where relevant.

Operating Model

- Plan the work from documented requirements and trace it to Blueprint or Specifications.
- Implement in small increments with tests, observability, and rollback behavior.
- Review changes against standards, threat model, failure modes, and acceptance evidence.
- Release only when documentation, tests, and operational notes are current.

Inputs

- Architecture Library documents, implementation tickets, configuration, schemas, policies, runtime context, and stakeholder constraints.
- Prior release notes, known risks, incident learnings, test results, and acceptance criteria.

Outputs

- Updated documentation, implementation plans, review evidence, test reports, operational runbooks, and release artifacts.
- Traceable decisions that explain why the selected approach is correct for Mariam.

Interfaces

- Git repository, CI/CD pipeline, runtime services, event bus, storage, dashboards, issue tracker, and release packaging.

- Human interfaces: review boards, chief office, department leads, operators, testers, and governance approvers.

Security

- Apply least privilege, secret isolation, signed releases where applicable, and audit-ready evidence.
- Do not expose tenant data, credentials, customer materials, private model artifacts, or privileged runtime outputs.
- Escalate security-sensitive deviations before implementation or release.

Metrics

- Lead time, review time, test pass rate, defect leakage, incident count, recovery time, documentation freshness, and acceptance completion.
- Topic-specific service metrics must be linked to dashboards and release notes.

Testing

- Verify expected behavior, failure behavior, recovery behavior, and evidence generation.
- Include automated tests where implementation exists and manual acceptance checks where documentation is the deliverable.
- Record test gaps explicitly in release notes.

Acceptance Criteria

- The workflow is clear enough for a new team member to execute without oral context.
- Required inputs, outputs, owners, interfaces, tests, and evidence are documented.
- Risks and security considerations are visible before release.
- The document is linked from INDEX, TRACEABILITY, ACCEPTANCE, MANIFEST, and release notes.

Traceability

Source	Relationship
MAL-V18-REVIEW-LOOPS-02	Governs Review Loops.
Volume 05	Engineering standards reference.
RELEASE_NOTES_v91_draft.md	Release evidence.

Version History

Version	Date	Status	Notes
v91 draft	2026-06-29	Draft	Added official documentation for Review Loops.